

Facial Recognition Bias Audit Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 35-45 min

Category	Knowledge and Technology	Time	35-45 min
Difficulty	Medium	ID	facial-recognition-bias-audit

Big idea: Automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

One-Minute Teacher Brief

Students audit a fictional facial recognition system and decide what evidence would justify trusting or limiting it.

Student Prompt

Inspect Facial Recognition Bias Audit Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Facial Recognition Bias Audit Stimulus A	Student-facing stimulus 1: Audit row: hallway camera, 94% overall accuracy, but higher false positives in low light. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Facial Recognition Bias Audit Stimulus B	Student-facing stimulus 2: Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Facial Recognition Bias Audit Reveal / Twist	Student-facing stimulus 3: Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Facial Recognition Bias Audit Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Facial Recognition Bias Audit Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	When is a technological classification reliable enough to act on?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how automated recognition is a knowledge claim whose reliability can vary across contexts and groups.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: When is a technological classification reliable enough to act on?

Setup Checklist

- Use fictional data and avoid using real student faces or sensitive biometric tools.
- Present the system as a claim-making tool: ‘this face matches this identity.’

Example Stimuli or Materials

- Audit row: hallway camera, 94% overall accuracy, but higher false positives in low light.
- Stakeholder card: student wrongly flagged, administrator managing safety, developer defending the model.

Resources To Use

- Resource pack: <resources/facial-recognition-bias-audit/index.html>
- Card deck CSV: <resources/facial-recognition-bias-audit/cards.csv>
- Visual prompt: <resources/facial-recognition-bias-audit/visual-prompt.svg>
- Audit table with context, group, false positives, false negatives, and decision risk.
- Deployment verdict sheet: use, restrict, revise, reject, and evidence still needed.

Run Sequence

1. Introduce a fictional school security tool and its claimed accuracy.
2. Groups inspect audit data broken down by context and subgroup.
3. Students identify where the headline accuracy hides reliability differences.
4. Role groups decide whether to deploy, revise, restrict, or reject the tool.
5. Debrief how technical evidence and ethical responsibility interact.

Debrief Moves

- Knowledge question: When is a technological classification reliable enough to act on?
- Can a system be accurate overall and still unjustified in practice?
- What evidence is needed before automated knowledge claims affect people?

Teacher Quick Script

- Overall accuracy is not the whole knowledge claim.
- Who bears the cost when a technological knower is wrong?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...